

How VIRAM Operates

VIRAM is truth-first verification middleware. It sits between an organisation's documents and any AI model, checks every answer against those documents, and corrects what fails.

Most AI systems work like this:

Files → Model → Answer

After what can sometimes be a few prompts on large files, the model starts fabricating. There is no warning. Nothing checks the answer against the files. The user is none the wiser.

VIRAM works like this:

**Files → Structure → Label → Retrieve only what's relevant →
Model reasons → Verify against files → Correct → Answer**

What that means in practice:

- An organisation's files are ingested once, without the model. Example: on a seven-volume, 1.3-million-word test corpus, ingestion accuracy measured 99.9%+ at word level, with 98%+ exact-match accuracy across 21,275 table cells — some of them in rotated, side-on tables.
- Per prompt, the system retrieves only what is relevant. A narrow prompt — finding contradictions, for example — can reduce 1.3 million words to a few thousand tokens (a few pages of text); broader prompts send up to around 100k tokens (roughly 75,000 words) to the model. The model never searches or holds the corpus. It does only the reasoning, which is what it is best at.
- Every claim in the model's answer is verified against the original files — file, section, paragraph and page. Anything that cannot be verified is removed, corrected, or flagged in a Verification Notice at the bottom of the output.
- Where the files do not contain the answer, VIRAM declines rather than fabricates — and tells the user so directly in the response.
- Nothing relevant to the prompt is silently dropped. If the relevant material exceeds what the model can take in one pass, the user is told. Content is only condensed when the prompt itself calls for it — summarise, evaluate, comment on flow.

The standard is faithful representation with provenance. The truth is the organisation's files — every answer is checked against them, and anything that cannot be verified is flagged, never hidden.

VIRAM draws answers only from the organisation's own files — never from the internet or the model's own data. If a model introduces its own content (rare, but possible), it is caught, corrected or flagged.

VIRAM is model-agnostic and can be deployed in the cloud, on-premises, or fully air-gapped for sovereign environments. It is built for governments and regulated institutions — anywhere documents carry legal, financial or public consequence.

The same architecture powers two companion products: IPIQ, which searches an organisation's own files for new knowledge it didn't know it held, and IIQ, a science goal exploration engine.

viram.ai